

MENOMINI QUANTITY*

David Pesetsky

The phonology of Menomini, an Algonquian language spoken formerly in northern Michigan and Wisconsin, has been outlined in an early paper and a posthumous grammar by Bloomfield (1939, 1962), and has been discussed from a generative point of view by Bever (1967). Like many Algonquian languages, Menomini shows a complicated alternating pattern of vowel length. This phenomenon has been discussed also by Howard (1972, pp. 138 ff.). In this paper, I hope to show that a number of odd features of the rules governing these alternations can be explained rather naturally in a metrical framework, as developed by Liberman & Prince (1977) and extended in recent work by Halle & Vergnaud (1978), and others.

Menomini shows an underlying distinction between long and short vowels, which is neutralized in the following cases, as described by Bloomfield (1939).¹

- 1)a In monosyllables, short vowels are replaced by long.

pe?c → pee?c 'by accident'

- b After a closed syllable, a long vowel in an open syllable is replaced by a short.

keeh kaa mew → keeh ka mew 'he berates him'

but keeh kaah tan 'he berates it'

- c If the first two vowels of a word are short, the second is replaced by a long.

ne ka naaw → ne kaa naaw 'he is left'

- d If the even (second, fourth, etc.) syllable after the next preceding long vowel ... is open and has a long vowel, this long vowel is replaced by a short.

a cee ce kaa po we heew → a cee ce ka po we heew

'he stands him upside down'

- e If the even syllable ... is closed and contains a short vowel, this short vowel is replaced by a long.

kee me wan ken → kee me waah ken (n → h)

'whenever it rains'

As Howard notes, the surface definition of "closed" and "open" syllable is rather unusual in Menomini. While the first syllable in a sequence ... (C)VC CV ... is predictably considered closed (cf. (1e)), a syllable (C)VC in word-final

position is considered open. Thus, the following changes occur:

- 2)a ko? taan → ko? tan 'it is feared' by (1b)
 b paa pe me ke heew → paa pe me ke hew 'he brings him up' by (1d)
 c na yooh ta wak -X→ *na yooh ta waak 'when I heard him' by (1e)

In a segmental treatment of these processes, these data present no particular problem, since one can write closed syllable rules to apply to vowels only in the environment /_CC, and open syllable rules to apply to vowels in the environment /_C {V}#. "Open"

and "closed" in such a theory are mere labels of convenience. We will be treating these processes, however, as dependent on a hierarchical syllable structure, in which "open" and "closed" will have a structural definition. Clearly, such a treatment cannot easily distinguish a (C)VC syllable word-internally from a (C)VC syllable word-finally and a (C)V syllable anywhere. The latter two do not form a structural natural class.

Fortunately, an examination of Bloomfield (1962, 4.35) suggests another treatment of final (C)VC syllables. Bloomfield notes the necessity of a rule of FINAL VOWEL DROP:

3) V Ø /_#

This rule will account for alternations like kaakek 'forever' ~ kaakekee-(a)hkamek (a → Ø) 'for all time'. Further evidence for Bloomfield's rule is supplied elsewhere in his grammar, where he observes that "where a syllabic is heard at the end of a word, this seems to be due, perhaps in all cases, to the dropping of a final h, w, or y...." (1.6)

Bloomfield does not suggest that all words underlyingly terminate in a vowel. If we assume such a word-structure constraint, however, and order (3) after closed-syllable rules in 1), surface-final consonants will no longer underlyingly close surface-final syllables. We will be able to regard surface (C)VC# as underlying (C)V CV#, and will thus explain why surface (C)VC# acts as an open syllable. The independently necessary FINAL VOWEL DROP rule will eliminate the final vowel. In many cases, it seems, examination of paradigmatic alternations can tell us what sort of final vowel has been dropped in a given word. Since I have not made such a study, I will represent final vowels with the symbol &.

A SEGMENTAL SOLUTION

The processes described in 1) have been treated adequately with segmental rules by Bever and Howard. Nonetheless, they have certain peculiar properties which are not explained in such an approach.

First, the processes of 1) do not apply to all words of Menomini, but only to so-called "mobile words". "Static

words" (some pronouns, particles, names, verbs, etc.) fail to undergo any of the processes of 1). In other words, if a rule fails to undergo any process in 1), it fails to undergo all the processes of 1). There is no natural explanation for this fact in a segmental approach. As we shall see below, some, but not all of the processes of 1) can be collapsed into unified segmental rules without extensive and uninformative use of conventions like angled brackets. Thus, the only way standard theory can capture the common failure of all the processes in 1) in static words is by a redundancy rule governing the assignment of rule features. This will link a negative value for one rule of 1) to negative values for all rules of 1). While such redundancy rules may sometimes be necessary (cf. the treatment of foreign words in Russian; Lightner, 1972), they do not offer an explanation of the linkages created. We shall see that an explanation is available in a metrical treatment.

Second, there is a curious overlapping in structural changes and environments among the processes of 1), which one wants to capture in a formal treatment. It is a simple matter to use curly brackets to collapse 1d) and 1e), where two changes take place in the same even-syllable environment; call this the EVEN SYLLABLE RULE (ESR):²

$$4) \left\{ \begin{array}{l} [V_{+long}]CV \rightarrow [V_{-long}]CV \\ [V_{-long}]CC \rightarrow [V_{+long}]CC \end{array} \right\} / [V_{+long}]C_0 [V_{-long}]C_0 (VC_0 [V_{-long}]C_0)^* \underline{\quad}$$

(cf. Howard, (36-7))

Now notice that 1b) shares its structural change with the upper line of the ESR 4), but with a different conditioning environment. One would like somehow to collapse the two rules. An obvious problem is created by ordering. 1b), as informally stated above, must precede 1c). This is because 1c) represents a surface true generalization: one of the first two syllables of a mobile word is always long. If 1b) were to follow 1c), it would create violations of this generalization:

- 5) UNDERLYING: ah koo k& 'the other comes next after him'
- 1b) o
1c) oo
3) ah kook
- but 1c) _____
1b) o
3) *ah kok

1c), in turn, must precede the ESR of 4):

- 6) UNDERLYING: a ce te kaa po we hee w& 'he stands him upside down'
- 1c) ee
4) a
3) a cee te ka po we heew

- but 4) _____
 1c) ee
 3)*a cee te kaa po we heew

As long as 1c) must be ordered between 1b) and the ESR, the latter two rules cannot be collapsed. 1c) can be ordered after the ESR, however, if we complicated the ESR's environment. 1c) guarantees that, if the first syllable of a word is not long, the second will be lengthened. As Bever notes, this means that the ESR will always apply either in even syllables counting from an initial long syllable, or it will apply in even syllables counting from the beginning of the word, i.e., as in 7), where capital S represents a potential target syllable for the ESR:

- 7)a $\left[\begin{smallmatrix} V \\ +long \end{smallmatrix} \right] s S s S s S s S s S s S s \dots$
 b $\left[\begin{smallmatrix} V \\ -long \end{smallmatrix} \right] S s S s S s S s S s S s S s \dots$

Regardless of what happens to the second syllable of 7b) under the ESR, it will be lengthened by 1c). If we complicate 4) as in 8a) below, we can order 1c), restated as 8b), after it:

- 8)a $\left\{ \begin{array}{l} \left[\begin{smallmatrix} V \\ +long \end{smallmatrix} \right] CV \rightarrow \left[\begin{smallmatrix} V \\ -long \end{smallmatrix} \right] CV \\ \left[\begin{smallmatrix} V \\ -long \end{smallmatrix} \right] CC \rightarrow \left[\begin{smallmatrix} V \\ +long \end{smallmatrix} \right] CC \end{array} \right\} \left\{ \begin{array}{l} \# \\ \left[\begin{smallmatrix} V \\ +long \end{smallmatrix} \right] \end{array} \right\} C_0 \left[\begin{smallmatrix} V \\ -long \end{smallmatrix} \right] C_0 (VC_0 \left[\begin{smallmatrix} V \\ -long \end{smallmatrix} \right] C_0)^* _$
 b $v \rightarrow [+long] / \# C_0 \left[\begin{smallmatrix} V \\ -long \end{smallmatrix} \right] C_0 _$

1b) now immediately precedes the ESR, and we can collapse the two rules. However, since 1b) must precede the ESR ---

- 9) UNDERLYING: keeh kaa mee w& 'he berates it'

- 1b) a
 8a) e
 3) keeh ka mew
 but 8a) _____
 1b) a
 3) *keeh ka meew

---and since it only shares the upper structural change with 8a), we must use the full arsenal of SPE bracketing conventions to effect the collapse. Believe it or not, we rewrite 1b) and 8a) as 10):

- 10) $\left\{ \begin{array}{l} \left\langle \left[\begin{smallmatrix} V \\ +long \end{smallmatrix} \right] CV \rightarrow \left[\begin{smallmatrix} V \\ -long \end{smallmatrix} \right] CV \right\rangle_a \\ \left[\begin{smallmatrix} V \\ -long \end{smallmatrix} \right] CC \rightarrow \left[\begin{smallmatrix} V \\ +long \end{smallmatrix} \right] CC \end{array} \right\} \left\{ \begin{array}{l} \langle VCC \rangle_a \\ \left\{ \begin{array}{l} \# \\ \left[\begin{smallmatrix} V \\ +long \end{smallmatrix} \right] \end{array} \right\} C_0 \left[\begin{smallmatrix} V \\ -long \end{smallmatrix} \right] C_0 (VC_0 \left[\begin{smallmatrix} V \\ -long \end{smallmatrix} \right] C_0)^* \end{array} \right\} _$
 Cond: if a then b

Thus, it is clear that a segmental treatment can collapse lb), ld) and le), although la) and lc) do not seem collapsable with these rules. It is equally clear, however, that such a collapsing provides no explanation of the peculiar features of these processes. For example, (A) What do the shortening of open syllables and the lengthening of open syllables have in common, that they should apply in the same environment? (B) Why should lb), the upper environment expansion of l0), condition a common structural change with only the open syllable part of the alternating length rule? In other words, why aren't closed syllables lengthened after closed syllables, since open syllables are shortened? Finally, (C) why is an alternating pattern such as that created by the lower environment expansion of l0) a reasonably natural, common phenomenon; i.e., why should a rule apply in alternate syllables, and not, say, in every fourth or fifth?

By permitting an albeit baroque collapsing of lb), ld) and le), the SPE formalism is telling us something important, namely that rules like these are valued, because they can be collapsed. In the following section, I will show that a metrical approach permits essentially the same sort of collapsing, but additionally offers explanations for the observations made above. In general, a metrical solution might not answer our questions; for example, it might be the case that some pseudo-Menomini applies lb) so that it will also lengthen closed syllables. We can show, however, that a metrical solution at least marks such a rule as less natural than the actual Menomini rule, and will permit us to exclude such a rule altogether, if such an exclusion should prove justified. The necessary constraint is well motivated and extremely simple.

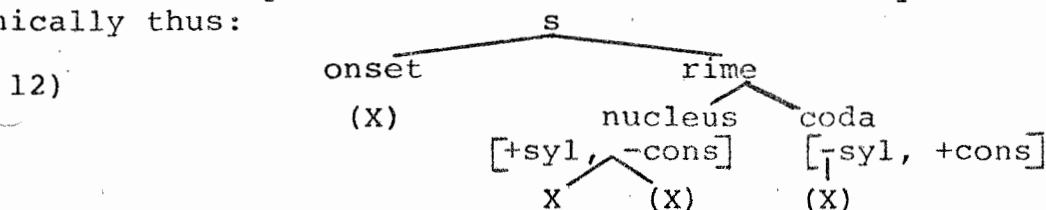
A METRICAL SOLUTION

Let us consider the alternating processes expressed by ld, e), which we have called the EVEN SYLLABLE RULE. As we have seen, these rules condition two different changes in an identical environment. Menomini syllables (forgetting about onset glides, which do not concern us) can be open or closed, long or short. Representing long vowels and diphthongs as we have, by sequences of vowels, we have the following four types:

11)a	(C)V	c	(C)VC
b	(C)VV	d	(C)VVC

In the environments of the ESR, only types a) and d) are permitted. Syllables of type b) or c) are altered to conform to this pattern, either by doubling a V or by deleting a V. A final C is not deleted.

We can represent the canonical Menomini syllable hierarchically thus:



As has been noted in recent work (Halle, MIT lectures, 1979), 15) represents an important and useful constraint on tree construction. It will play an essential role in our analysis.

Note that 15) is true relative to the projection under which a rule of tree construction is operating. For example, if trees are constructed on a nucleus projection, two syllables with branching nuclei (long vowels, etc.) cannot be footmates, since this would create a tree both left- and right-branching. On the other hand, nothing will prevent two syllables with branching rimes (closed syllables) from sharing a foot, since on the nucleus projection the rime node is ignored. Note further that the notion of "branchedness relative to a projection" obeys a kind of subjacency condition. Rules of tree construction will only take into account the branchedness of the nodes being projected. Lower levels of structure are not considered. This is essential if we are to use, for example, the distinction between branching and non-branching rimes in Menomini to capture the distinction between closed and open syllables. An open long syllable will have a branching nucleus, but the syllable must still count as non-branching on the rime projection.³

Returning to the ESR, we have a simple means of characterizing this rule, which defines the environments in which condition 14) is activated. The simplest formulation of the rule will precede 1c).⁴ Recall that an ESR which precedes 1c) will need to single out (A) the even syllable from the last long vowel (cf. 7a, b) above) and (B) the even syllable from the beginning of the word.

This can be done easily. Assume a left-to-right process of binary bounded foot formation, subject to the following restriction on initial node n1 and final node n2 in each foot:

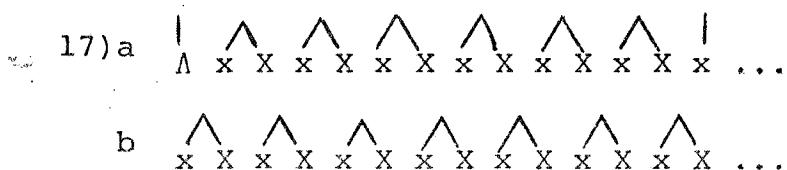
- 16)a On the nucleus projection, n1 cannot branch.

Given constraint 15), we can restate this condition as follows:

- 16)b On the nucleus projection, feet cannot be left-branching.

In other words, the left branch of each foot constructed by this rule may not dominate a long vowel, which is a branching nucleus.

Condition 14) applies to the right branch of each foot so constructed. Thus, the capital X's below correspond to the capital S's in 7a, b), and indicate the syllables to which 14) is applicable:



The first symbol in 17a) represents a branching node on the nucleus projection, i.e. a long vowel. By 16), it is

Thus, we assume that the ESR is characterized as follows:

20) ESR (Metrical)

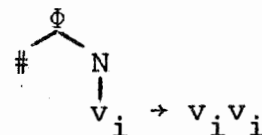
- a Project nuclei.
- b Left to right, construct binary bounded feet, n1 cannot branch (i.e., feet cannot be left-branching)
- c Condition 14) must be met by the syllable contained by the right branch of each foot, either by deletion or copying under identity.

We must note an obvious and interesting problem with this formulation. Foot construction in the ESR is sensitive only to the branching structure of syllabic nuclei. Nonetheless, condition 14), which applies to the right branch of these feet, is sensitive to the structure of the entire rime. (It thus also violates the "subjacency condition" discussed above with regard to foot construction rules, since it must analyze all branching structure contained by the rime node.) This suggests that the projection used in tree construction and the projection used for subsequent operations on these trees may differ. I will not try here to formulate what is lacking: an explicit theory of projections which will spell out how they work and interact with other systems (but see the Appendix to this paper). The observation is important, however, and shows that this analysis somewhat contradicts the spirit in which the mechanism of projection has been used (cf. other papers, this volume). I will not return to this problem in the body of this paper.

Now let us turn to rule 1c), which we will call SECOND SYLLABLE LENGTHENING (SSL). This can now be seen simply as a lengthening rule in the right branch of the first foot. If the first syllable is long, it will automatically be a degenerate foot by itself, and the rule will not apply. We will want to revise this formulation later. For now, we can state SSL thus:

21) SSL

In the structure created by the ESR,



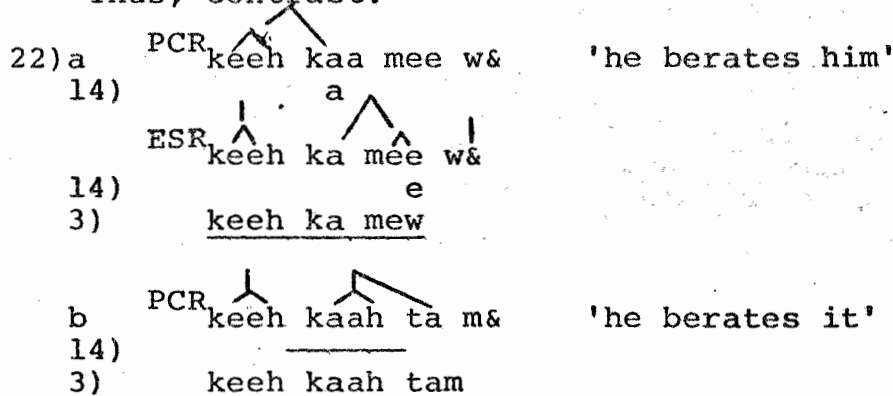
The working of this rule was demonstrated in 18a) above. ⁵
As mentioned, it follows the ESR.

Now let us look at 1b), which we will call the POST-CLOSED RULE (PCR). As we saw in 9), this must precede the ESR. Recall that the PCR says that following a closed syllable, 14) must be met by open syllables, but not by closed syllables. In the segmental treatment of this process, this was an inexplicable fact which had to be written into the collapsed version of the rule with angled brackets and indices. In contrast, this fact has an elegant explanation within metrical theory as developed here and elsewhere in this volume.

Recall that we imposed a general condition on trees in 15), which required that in a binary branching tree one node be prohibited from branching, i.e., that a tree not be both left- and right-branching. Let us allow a rule-specific option of requiring a node to branch, i.e., of requiring a tree to be obligatorily left- or right-branching. This is a necessary option for any phonological process that (A) is triggered by certain designated structures but (B) applies to non-designated structures. For example, any stress rule which scans a string for heavy or long syllables to initiate tree construction (e.g., E. Cheremis, Mongolian, etc.; see Halle & Vergnaud) will use this option.

We will assume that the PCR in Menomini uses this option. The PCR tree construction procedure constructs binary bounded feet on the rime projection (which distinguishes closed from open syllables) with the condition that such feet be obligatorily left-branching, i.e., n1 must branch. Thus, to construct trees for the PCR, one scans a string of rimes; when one finds one which branches, i.e., a closed syllable, it will form the left branch of a foot, if possible. The right branch will be subject to l4). Crucially, condition l5) ensures that the right branch of a foot will never contain a closed syllable under this rule. Since the left branch must be a branching node, the right branch cannot be branching. Thus, the fact that the PCR only imposes a phonological change on open syllables under l4) is an immediate consequence of the nature of the triggering structure for foot formation (branching rimes) and the independently motivated universal condition l5). Nothing extra need be said.

Thus, contrast:



The syllable kaah in 22b), since it is branching, cannot be the right branch of a tree created by the POST-CLOSED RULE. This is a direct consequence of general principle 15).

Note also that the non-iterative character of the PCR, as opposed to the ESR, also follows directly from the necessary statement of the rule and from principle 15). The ESR is "iterative" because no positive conditions need be put on each foot. The rule scans a string from left to right. Any node which does not branch can be a left branch of a foot, and the following node can always be adjoined as the right branch. There is no special trigger for foot construction. The PCR, by contrast, appears "non-iterative" because a positive condition is put on the left branch of each foot, requiring it

to branch. Condition 14) applies to the right branch of feet created by both rules. Since the special condition on the left branch of feet under the PCR is independently necessary to express the environment of the rule, it is an advantage of the metrical framework that the non-iterative character of the rule follows automatically.

Thus, we propose to state the PCR as follows:

23) PCR

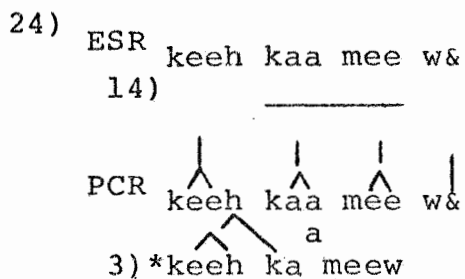
- a Project rimes.
- b Construct binary bounded feet, nl must branch (i.e., feet must be left-branching).
- c Condition 14) must be met by the syllable contained by the right branch of each foot, either by deletion or copying under identity.

The derivation of 22a) shows that we must assume that old tree structure is erased after the PCR. In other words, we must add 23d):

23d) Erase tree structure.

At our present state of knowledge, we must write this into the rule, since it is not clear when such erasure takes place in metrical rules. It does not apply after every rule of tree construction. For example, the rule of SECOND SYLLABLE LENGTHENING (SSL) discussed above must have access to the tree structure created by the ESR. More dramatically, in rules governing stress assignment, a "word tree" may be built on top of previously constructed foot trees.⁵

Finally, it should be observed that the ordering of the PCR before the ESR is crucial. The shortening of syllables under the PCR may permit such syllables to constitute left feet under the ESR. Contrast the bad derivation 24) with 22a):



Thus, in a sense, the two rules apply in feeding order.

Earlier, we asked three questions about Menomini quantity alternations and about such rules in general, which we noted could not be answered by segmental phonology. Two of them have been explained: (A) What is the common feature of the shortening of an open syllable and the lengthening of a closed syllable? (Maximality and minimality of branching.); and (B) Why does the POST-CLOSED RULE share only the change

affecting open syllables with the EVEN-SYLLABLE RULE? (Condition 15)). We have also shown how to extract the common feature of the PCR and the ESR, by generalizing condition 14) and showing that it applies to right branches of feet created by either rule. The condition is blind to which rule has created a given root, but merely applies upon construction of it. The condition is met by a minimal change in the structure of the affected rime.

However, we have not answered a deeper question we posed---namely, (C) Why should alternating rules, singling out every other syllable under certain conditions, be natural, while rules which might, for example, single out every fourth syllable or every seventh are unnatural. We will return to this question, in a tentative manner, only in the appendix.

I want to conclude this paper by noting that the rules proposed here, with a minor modification, can account for an area of Menomini phonology that has been problematic in previous treatments. This is the problem of so-called "glottal words". These are words whose first syllable has the form (C)V?, where the vowel is always short. These words do not undergo the SECOND SYLLABLE LENGTHENING (SSL) rule, which we wrote above as 21). This phenomenon creates problems elsewhere in a system like Bever's, in which, despite the observation credited to him above (cf. 7a, b)), the ESR applies after SSL and counts syllables from a long vowel. In a glottal word, there is no long vowel in one of the first two syllables to count from. The phenomenon does not, however, create any problems for an account like ours, in which the ESR precedes SSL (and cf. Howard). The ESR, grouping nodes into binary feet from the beginning of the word, will apply as we have stated it. We need merely find some way of blocking SSL from applying in glottal words.

An answer may be found in Menomini stress, which is, unfortunately, rather sketchily and confusingly described by Bloomfield (1962).⁶ Bloomfield seems to intend the following rules to apply (1.17; 1.49-53):

- 25)a Primary stress falls on a long vowel or diphthong in the penult syllable.

e nee niw 'man'

os kaa pee wes 'ceremonial attendant'

see peew 'river'

- b Primary stress also falls on every long vowel or diphthong followed in the next syllable by a short vowel.

mee we ne se he wak 'they drive him away'

- c Secondary stress falls on even-numbered vowels in a succession of short syllables (1.17; contradicted at 1.51), but not on a syllable in word-final position.

mée we ñě se hẽ wak 'they drive him away'

keh kẽ men new

The metrical theory of stress does not allow for the possibility of more than one primary stress per stress group. Reading between the lines of Bloomfield's description, one might assign stress in the following fashion:

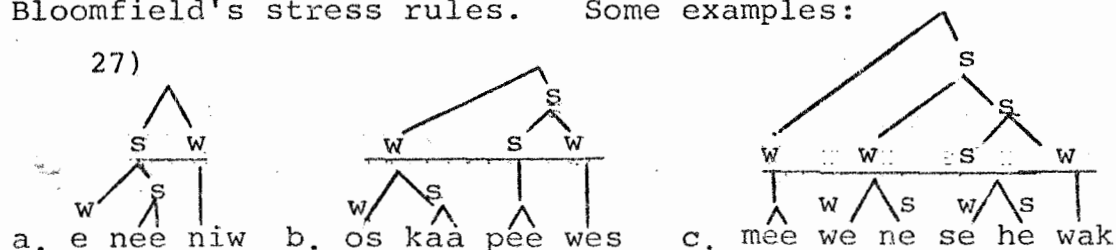
Let us assume that the basic rhythmic pattern of Menomini, as 25c) suggests, is the same as the foot structure created by the ESR. Let us assume that nodes of these feet are labelled by convention LIB (Left strong iff it branches). This will yield an essentially iambic pattern beginning with the syllable after a long vowel. Over these feet we will construct a word tree. 25a) suggests that this tree will be right branching and be labelled by convention RIB (Right strong iff it branches). This will assign stress to the penultimate foot. The stress in seepew above shows that this must be done after the rule of FINAL VOWEL DROP 3):



Penult foot stress will place primary stress on any long vowel or diphthong in the penult syllable; it will also stress a short vowel in penult position, and will, for example, place main stress on he in meewenesehewak of 1b, c). (This last is specifically contradicted by Bloomfield, who places a main stress on the initial, long vowel in this word. It may be the case, however, that in the mapping from metrical trees into phonetic stress, a special rule gives stress to long syllables immediately dominated by s. This might also account for 25b).)

Penult word stress will also account for the observation that a vowel in word-final position never receives secondary stress. (A short vowel in that position is mentioned by Bloomfield; a long vowel in that position will not receive stress by his rules, since it is neither penultimate nor followed by a short vowel.) Such a vowel is in the most embedded foot of the tree. If any foot will receive the least prominence, it will be that one.

Such, then is an attempt to make metrical sense of Bloomfield's stress rules. Some examples:

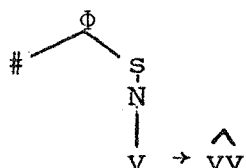


This fragmentary account of Menomini stress has been brought forward because of its relation to the problem of

glottal words. Bloomfield notes that in odd-numbered syllables of a sequence of short-vowel syllables, a process of vowel reduction takes place. These are, of course, weak left-branch positions. In addition, an initial short vowel is reduced to the point where oppositions of height are neutralized, and vowels are distinguished only by frontness and backness. Crucially, neither of these processes apply in syllables of the form (C)V?. We can account for these peculiarities if we say that reduction processes apply to syllables dominated by w at the foot level, and that (C)V? syllables are idiosyncratically marked s. If we rewrite the SECOND SYLLABLE LENGTHENING rule so that lengthening applies, not to any first foot right branch, but to a strong first foot right branch, we will have an account of the exceptionality of glottal words with respect to this rule:

28) SSL (revised)

In the structure created by the EVEN SYLLABLE RULE,



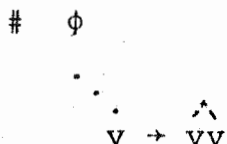
The labelling of glottal-final syllables with s is an ad hoc move, yet it is one that allows a number of facts to be captured rather easily (see Bever for another treatment). Clearly, we do not have a satisfactory resolution of the problem. Glottal-final syllables have a number of the properties of long syllables---resisting vowel reduction, counting as long for a vowel raising rule discussed in the authors cited. This, combined with the fact that the final consonant in a V?C# output of FINAL SYLLABLE DROP 3) does not have to truncate, suggests that /?/ may be part of a syllabic nucleus, and not part of a coda. However, for other processes, like the POST-CLOSED RULE, the EVEN SYLLABLE RULE, and the attendant condition 14), syllables with /?/ act as if /?/ were part of the coda, and not part of the nucleus, e.g.:

- 29) PCR  ko? taa n& 'it is feared'⁸
 14) a
 other ko? tan
 rules + 3)

One more complication of the revised SSL of 28) also allows us to capture 1a), which we have not so far discussed. This is the lengthening of monosyllables. Monosyllables will, of course, be neither a right nor a left branch of a foot, nor will they be dominated by any relative value label, i.e., by s or w. We can unify the monosyllable rule with the SSL then as follows:

30) SSL

In the structure created by the EVEN SYLLABLE RULE,



where $\underline{v}$ is not dominated by the left branch of ϕ and v is not immediately dominated by w .

This rule will lengthen vowels in three cases:

(1) monosyllables, (2) strong right branch vowels, and (3) initial feet dominating only one node. (3) includes only syllables which are long to begin with, and cannot be left-branches of ESR feet, so it is an irrelevant case.

Finally, a note on the difference between "mobile" and "static" words. We want to be able to explain the fact that a static word is an exception to all the processes described here (except, perhaps, the final vowel drop rule 3)). An explanation might be that the syllables of static words are lexically marked opaque to analysis by metrical rules. From this it will automatically follow that condition 14) and the SSL rule above cannot apply, since they will not be able to analyze infra-syllabic structure. It may be that rules of tree construction and labelling are able to analyze the syllables of such words, since there is no indication that they receive exceptional stress. Bloomfield's descriptions here do not yield sufficient information for an analysis to be well-motivated. It is obvious, however, that a metrical approach allows a single stipulation to exclude all static words from quantity alternations. We have seen that a segmental approach cannot do this so easily.

In this paper, I have suggested that, although segmental and metrical accounts of Menomini quantity alternations can be of equal empirical adequacy, a metrical account offers explanations of the surface properties of the phenomena involved which are not forthcoming in a segmental treatment. To be sure, both accounts require a number of ad hoc moves --- for example, to handle glottal words ---, but a closer study of Menomini phonology may elucidate these problems.

ORDERING OF RULES

POST-CLOSED TREE CONSTRUCTION

CONDITION 14

Erase brackets

EVEN SYLL RULE TREE CONSTRUCTION

CONDITION 14

SECOND SYLL LENGTHENING (includes monosyllable rule)

APPENDIX

The purpose of this note is to suggest a possible approach to question (C) asked above, namely, why there should be rules singling out alternating syllables, but no rules singling out syllables separated by more than one entity. In metrical terms we might rephrase the question thus: we know that rules of tree construction must be allowed to create bounded binary and unbounded feet. Why are there no bounded quaternary, nonary, etc. feet? We assume with Hayes (this volume) that the device of extrametricality takes care of cases for which it has been thought that ternary feet are necessary (pace problems with English). We also assume that so-called "degenerate feet" which dominate only one node are not specified by tree-construction rules, but result from the interaction of these rules with universal conditions, as we shall see.

In this long appendix, I will not work out the details of this approach for the full range of interesting languages which have been brought into the discussion of metrical theory, but hope to do so in a later paper. I shall, however, show that the Menomini rules discussed above fall neatly within this theory, and help exemplify the restricted range of available options.

The theory itself is quite simple. I propose that the surface distinction between bounded binary and unbounded feet is an artifact of the interaction of two parameters for tree construction rules parameters which must be specified in any case. Each rule must have a direction of scan. This is the order in which projected entities are looked at for inclusion in a tree, either from left to right or from right to left. In addition, we have seen that we must know whether the trees constructed are to be left-branching or right-branching. In other words, we must know a rule's branchedness.

To see how the system works, we must be precise about the nature of tree construction. The designated projection for a given rule will provide an ordered set of entities, all of which must be scanned by the rule of tree construction. The designated scan tells us whether to begin at the beginning or the end of the word. Beginning at the appropriate end of the word, and scanning in the proper direction, each entity is examined by the rule and the following question is asked: Can this element form a branch of an already constructed foot? If the answer is positive, then this branch is formed. If the answer is negative, then the entity forms its own foot, which will be degenerate if no branches are added later in the scan.

It will be seen that this procedure, interacting with a designation of branchedness, will provide binary and unbounded feet iterating in either direction, but with certain restrictions on their type.

Consider, for example, a language with initial stress, like Hungarian. We want to construct an unbounded foot beginning with the initial syllable. Let us suppose that such a tree is constructed by a rule with the following parameters chosen:

Replacing all "left" with "right" and vice-versa in 31) will yield a rule assigning final stress (Halle & Vergnaud cite French and Farsi as examples). Reversing the labelling convention alone will yield second-syllable stress (Lakota and Araucanian are cited), while reversing everything but the labelling convention will yield penultimate stress (Polish, Quechua, etc.).

The interesting case is when we reverse only the branch-
edness or only the scan. Let us keep a left-to-right scan, but reverse the designated branching. (We will consider labelling shortly.)

37) Scan: L to R
Branching: R

Once more we examine a string of syllables, and suppose that no branching structure is projected (the simpler case):

38) $\begin{array}{ccccccc} | & | & | & | & | & | & | \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \dots$

The first two steps follow as in 33) and 34) above. An initially degenerate foot is formed on 1, and 2 can be attached as a right branch without creating an illegal left-branching structure:

39) $\begin{array}{ccccccc} & \diagup & & & & & \\ | & & | & | & | & | & | \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \dots$

We find a difference, however, when the scan reaches 3. We ask whether 3 can form a branch of a previously constructed foot, and find the answer to be no. Adjoining 3 to the previously constructed foot would create a left-branching structure as in 35), which is forbidden by the rule. Thus, 3 may not be adjoined, but forms a temporarily degenerate foot by itself:

40) $\begin{array}{ccccccc} & \diagup & & / & & & \\ | & & | & | & | & | & | \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \dots$

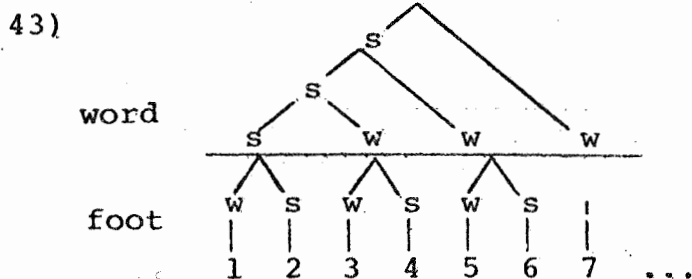
Proceeding to 4, we find that 4 can be a right branch of the foot dominating 3:

41) $\begin{array}{ccccccc} & \diagup & & \diagup & & & \\ | & & | & | & | & | & | \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \dots$

5, once more, cannot form a branch of a preceding foot. Thus, binary feet are formed from left to right as a natural result of the process of foot formation, the designated scan and the designated branchedness:

42) $\begin{array}{ccccccc} & \diagup & & \diagup & & \diagup & \\ | & & | & | & | & | & | \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \dots$

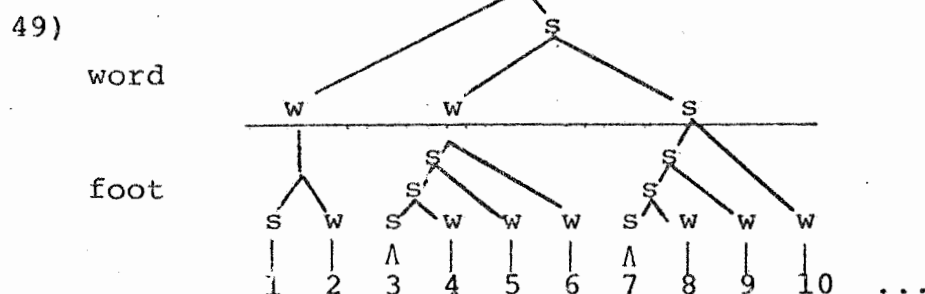
Let us suppose that the convention "R strong iff branching: applies to a tree constructed in this way, and the result is the foot level of Southern Paiute (cf. Halle & Vergnaud). In Southern Paiute, words show an iambic pattern of stress. This follows from a foot rule like 37) and a word-level rule assigning initial stress, i.e., 31) applying to the feet:



We have seen that the difference between bounded and unbounded feet can be made to fall out from other principles in simple cases where no scanned entities are projected as branching. Let us now see some more complex cases, where some entities are projected as branching. For unbounded feet, the familiar examples are Eastern Cheremis and Komi. Halle & Vergnaud's solution is immediately capturable in our system. The facts of Eastern Cheremis are:

- 45)a The accent falls on the syllable containing the last full vowel of the word.
- b If the word has only reduced vowels, the accent usually is on the first syllable.
- (H & V 6.18)

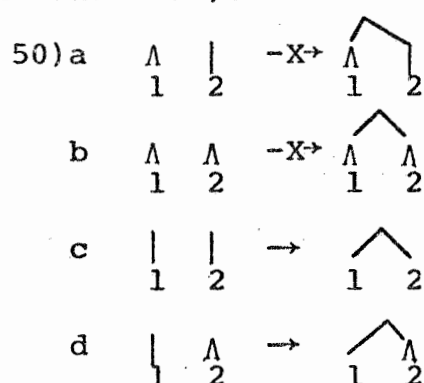
The labelling convention (R strong iff branching) applies, making the initial node of each foot the designated terminal element. Then a word-level tree using the mirror-image of 31) assigns stress to the final foot:



In words with no full vowel, initial stress will be assigned as in 36), at the foot level, and the word level will play no role.

Komi shows the opposite stress pattern, which is achieved by reversing the Cheremis rules.

For bounded feet, rules sensitive to branching structure are found in languages like Creek, Macushi Carib (Hawkins, 1951) and in the Menomini cases discussed above. Interestingly, each of these cases behaves as predicted by our theory. We have seen how a simple alternating pattern of binary feet is created by a rule like 37). Clearly, if some entities are projected as branching in this rule, no element can be attached as a right branch to a branching node, since the resulting tree would, illegally, be left branching. An entity which itself branches can, however, be attached to a non-branching node as a right branch, since right-branching structures are not excluded by the rule. In other words, if 2 below is the element being scanned, the following pattern of adjunctions will be found under 37):



In 50a), 2 will now be eligible to be a left branch of a following foot. In 50b), 2 will not be so eligible, and a succeeding entity, 3, will not be attachable to any foot formed by 2.

This is, of course, exactly the situation found in our EVEN SYLLABLE RULE in Menomini. In Creek, too, binary bounded feet are constructed left to right, subject to the condition that the left branch of such feet not branch. Main stress falls on the last foot so formed. (Long or closed syllables count as

branching in Creek---a problem for our "subjacency" constraint suggested above.) In Macushi Carib, the same rule applies in the same way, with vowels dominated by left branches getting deleted. (Interestingly, branchedness here seems to be determined by a branching rime or branching onset, if Hawkins' analysis is correct.)

In the text above, we did not rely on a general theory of tree construction like the one provided here, and we simply stipulated that, for the ESR in Menomini, the left branch of a foot cannot be itself branching. Without a general theory like the one proposed, one might suppose that a rule could construct binary feet from left to right with the opposite stipulation, namely that the right branch cannot be branching. In 19), we saw that this was untenable as an approach to the Menomini ESR. I do not know of a case in which such a rule is necessary at all. In the rule allowed by our system, in a word with an alternating pattern, a branching node in an odd syllable will disrupt the count. In the rule disallowed by our system, a branching node in an even syllable will disrupt the count. This latter case might be found on the surface in our system too, since extrametricality allows one "not to count" a syllable in first or last position in a word. Note, however, that the device of extrametricality must at least have a cost associated with it, rendering languages employing the device somewhat marked, thus accounting for the apparent rarity of the latter case.

We must allow one more option in our formulation of rules, as noted by Halle & Vergnaud and above. We must allow trees to be, not "optionally" right- or left-branching, as in the rules considered above, but also "obligatorily" right- or left-branching. Two examples of this option are provided, for unbounded feet, by Aquacatec (Halle & Vergnaud) and by our POST-CLOSED RULE of Menomini, for bounded binary feet.

As described by Halle & Vergnaud, Aquacatec, like Eastern Cheremis, assigns stress to the last long vowel of a word, but assigns stress to the last, not first, vowel in words without long vowels.

The foot-level rule for Aquacatec will be identical to that for Cheremis, but will have the following value for an additional parameter:

51) Obligatory: Yes

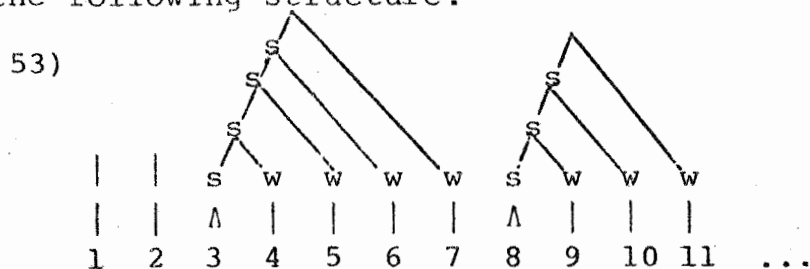
Thus, consider the sequence:

52)

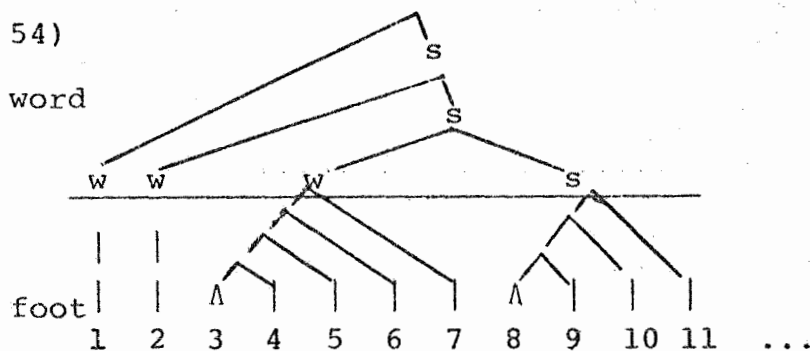
		Λ					Λ				
1	2	3	4	5	6	7	8	9	10	11	...

1 forms a degenerate foot. 2 cannot form a right branch of the foot dominating 1, since such a foot will not be explicitly left-branching. It similarly forms a degenerate foot by itself. (Note that degenerate foot formation is a given of UG, a part of the general process of tree construction, and is not subject to the "obligatorily left-branching" condition on the tree-

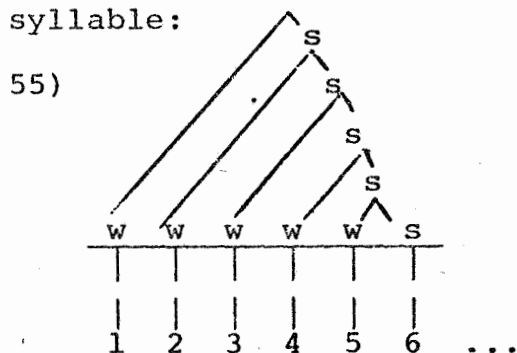
construction rule.) 3 once more cannot be adjoined to any previously constructed foot. 4, however, can be adjoined to 3, since the resulting foot will be left-branching. The end result is the following structure:



A word-level tree will assign stress, as in Eastern Che-remis, to the final foot.



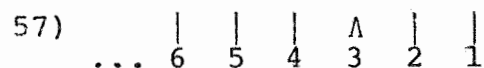
The obligatoriness condition will prevent any but degenerate feet from being constructed in a word without long vowels. The word tree will thus automatically stress the final syllable:



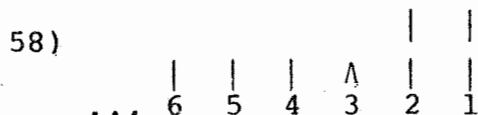
The POST-CLOSED RULE of Menomini can also be easily expressed within our system. We must suppose that this rule has a right to left scan, and is the mirror image of 37), plus an obligatoriness condition:

56) Scan: R to L
Branching: L
Obligatory: Yes

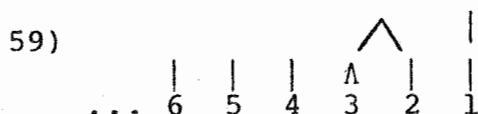
Consider a structure like:



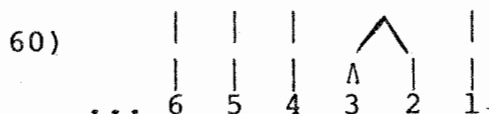
We look at 1. It cannot form the left branch of any foot previously constructed, so it becomes a degenerate foot. 2 cannot be adjoined to the foot created by 1, since the resulting foot would not be left-branching, so it too is dominated by a (temporarily) degenerate foot:



Now we scan 3. If joined as a left branch to the degenerate foot dominating 2, the resulting foot will be left-branching, and will satisfy the conditions on rule 56):



4 cannot be adjoined to the foot dominating 3, since this foot would be right-branching. Similarly, the obligatoriness condition on rule 56) prevents 5 and 6 from being adjoined to previously created feet. Thus, 4, 5 and 6 are all contained by degenerate feet:



Condition 14) then applies, as discussed, to the one right branch in the trees constructed.

To sum up, I wish to claim that the difference between unbounded and bounded feet, and the properties of the rules discussed above, all derive from the interactions of choices made by each tree construction rule among the following parameters:

61) Projection:	various feature bundles
Scan:	L to R/R to L
Branching:	L branching/R branching
Obligatory:	Yes/No
Labelling:	L strong iff branching/R strong iff branching

In a future paper I hope to show that this theory can handle various problematical cases in metrical theory.

FOOTNOTES

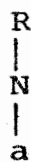
* Thanks to Morris Halle and Donca Steriade for discussion and enlightenment.

1. Also cf. Goddard, Hockett and Teeter (1972).

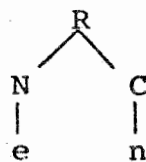
2. Rules like 4) can also be written as iterative rules (q.v. Howard). This issue does not bear on the general question of segmental vs. metrical rules. Howard also notes that the process suggests an alpha-rule with features $[\pm \text{long}]$ and $[\pm \text{closed}]$, but notes the weakening of feature theory attendant on the use of features like $[\pm \text{closed}]$.

3. Ken Safir (this volume) has shown a need for a similar subadjacency condition in Capanahua. Note that some way around this condition will have to be found for languages like English and many others, where "branchedness" takes into account both rime and nucleus branchings. In such languages, long syllables and closed syllables both count as "heavy":

e.g. páraffin



parental



coróna



I do not know whether the subadjacency condition is simply absent in these languages (and if so, whether absence or presence is the unmarked situation) or whether some other mechanism is at work.

4. This is because 1c), as a surface-true generalization, overrides changes required by 14). We could write an exception into 14), e.g. "except in the right branch of the first foot" (see below in text), but the superiority of the ordering advocated here is obvious.

5. This treatment of the interaction between the two rules of tree construction, the ESR and PCR, implies that it is no accident that no rules applying after the ESR must make reference to the PCR tree. Alternatively, one might suppose that there is no erasure convention as in 23d), but that different trees can be constructed on different projections and be referred to at any time in the course of a derivation. (This suggestion was made to me by Morris Halle.) This latter theory is conceptually simpler, but somewhat less restrictive. In the absence of deciding evidence the point is, in any case, moot. Note that any theory must crucially still order ESR tree construction after condition 14) applies to the PCR tree, as indicated in the following text paragraph.

6. The posthumous nature of the work removes blame from Bloomfield.

7. Bloomfield also indicates that there is some variation in disyllabic words, which I have not discussed here. And when all is said and done, he admits that "there is much fluctuation".

8. Actually, this could be derived by the ESR, were the PCR not obligatory. The point would still hold, though, since ko? could not be the left branch of an ESR foot if its nucleus were branching. Thus, ? must be in the rime by this test too.